

THE SUPPORT OF THE CHARACTER OF A SUPERCUSPIDAL REPRESENTATION

PIERRE DELIGNE

The character of a supercuspidal representation of a semisimple group over a local field is supported on the union of the compact open subgroups.

1. REVIEW OF MONOGENIC GROUPS

Let V be a finite-dimensional vector space over a field k , and let $g \in \mathrm{GL}(V)$. If k is perfect, the Zariski closure $\langle g \rangle$ of the group generated by g is the product of a multiplicative group $\langle g \rangle_m$ and a unipotent group $\langle g \rangle_{\mathrm{un}}$. If $g = su = us$ is the Jordan decomposition of g into its semisimple and unipotent parts, then

$$\langle g \rangle_m = \langle s \rangle, \quad \langle g \rangle_{\mathrm{un}} = \langle u \rangle.$$

Without any hypothesis on k , define the subgroup $\langle g \rangle_m$ of $\langle g \rangle$ as the Zariski closure of the union of the subgroups of $\langle g \rangle$ that are kernels of multiplication by n , where n is prime to the characteristic exponent of k . This definition is compatible with extension of scalars, and for k perfect it is equivalent to the preceding one. Over an algebraic closure $\bar{k}$ of k , the character group of $\langle g \rangle_m$ is the subgroup X of $\bar{k}^*$ generated by the eigenvalues of g , and

$$s \in \langle g \rangle_m(\bar{k}) = \mathrm{Hom}(X, \bar{k}^*)$$

corresponds to the identity inclusion of X .

Assume that k is a complete valued field, with uniformizer π . Let v be the valuation of k , with values in $\mathbf{Q}$, normalized by $v(\pi) = 1$, and let $n > 0$ be an integer such that $nv(X) \subset \mathbf{Z}$. The homomorphism $nv(\chi) : X \rightarrow \mathbf{Z}$ transposes to a homomorphism

$$m_g : \mathbf{G}_m \rightarrow \langle g \rangle_m.$$

For a linear algebraic group G over k and an element $g \in G(k)$, the preceding constructions may be repeated after choosing an embedding $G \subset \mathrm{GL}(V)$. The homomorphism $m_g : \mathbf{G}_m \rightarrow \langle g \rangle_m \subset G$ does not depend on the embedding, once n is fixed. For every linear representation W of G , it defines an action of $\mathbf{G}_m$ on W , equivalently a grading of W (SGA 3, I, 4.7.3). The part of degree d is the sum of the generalized eigenspaces of g corresponding to eigenvalues $\alpha \in \bar{k}$ such that $nv(\alpha) = d$.

Assume that G is reductive. Every homomorphism $m : \mathbf{G}_m \rightarrow G$ defines two opposite parabolic subgroups of G , namely the parabolics contracted

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and expanded by m . Let P_g^+ be the parabolic contracted by m_g , or by g . It has the following equivalent characterizations:

- (a) An element $x \in G(\bar{k})$ lies in P_g^+ , respectively in its unipotent radical U_g^+ , if and only if the morphism

$$\lambda \mapsto m_g(\lambda)xm_g(\lambda)^{-1} : \mathbf{G}_m \rightarrow G$$

extends to $\mathbf{G}_a$, respectively if this extension has limit e as $\lambda \rightarrow 0$. Equivalently, $\lim_{n \rightarrow \infty} g^n x g^{-n}$ exists, respectively equals e .

- (b) $\text{Lie } P_g^+$, respectively $\text{Lie } U_g^+$, is the sum in $\text{Lie } G$ of the generalized eigenspaces of $\text{ad } g$ whose eigenvalues have valuation ≥ 0 , respectively > 0 .

Put $P_g^- = P_{g^{-1}}^+$, the parabolic expanded by g and contracted by $m_{g^{-1}} = m_g^{-1}$, and let U_g^- be its unipotent radical. The parabolics P_g^+ and P_g^- are opposite, and put $M_g = P_g^+ \cap P_g^-$.

2. STATEMENT AND PROOF OF THE THEOREM

Assume that k is locally compact and that G is reductive. Let τ be the projection from G to its adjoint group G^{ad} . An admissible representation $(\mathcal{H}, \rho)$ of $G(k)$ is *supercuspidal* if its matrix coefficients are supported on inverse images of compact subsets of $G^{\text{ad}}(k)$.

If K is a compact subgroup of $G(k)$, let dk be the normalized Haar measure on K and let $[K]$ be its image in the algebra of compactly supported distributions on $G(k)$. The operator $\rho(K) = \int_K \rho([K])$ is the projector from $\mathcal{H}$ onto the subspace $\mathcal{H}^K$ of K -invariants. Saying that ρ is supercuspidal is equivalently saying that, for every compact open subgroup K , one has $\rho(K)\rho(g)\rho(K) = 0$ for $\tau(g) \in G^{\text{ad}}$ outside a suitable compact set depending on K . Let G_c^{ad} denote the union of the compact subgroups, or equivalently of the compact open subgroups, of $G^{\text{ad}}(k)$.

Theorem. *If $(\mathcal{H}, \rho)$ is an admissible supercuspidal representation of $G(k)$, then its character is supported on $\tau^{-1}(G_c^{\text{ad}})$.*

Choose local coordinates near e : a lattice $L \subset \text{Lie}(G)$ and an analytic isomorphism α from L to a neighborhood of e , such that

- (1) $\alpha(0) = e$, and the differential of α at 0 is the identity on $\text{Lie}(G)$.

Let $g \in G(k)$. We choose L so that

- (2) $L = (L \cap \text{Lie } U_g^+) \oplus (L \cap \text{Lie } M_g) \oplus (L \cap \text{Lie } U_g^-)$. Write this as $L^+ \oplus L^0 \oplus L^-$.

- (3) $\text{ad } g(L^+) \subset L^+$, $\text{ad } g(L^0) = L^0$, and $\text{ad } g^{-1}(L^-) \subset L^-$.

Since P_g^+ and P_g^- are stable under $\text{ad } g$, condition (3) may also be written as

- (3') $\text{ad } g(L \cap \text{Lie } P_g^+) \subset L$ and $\text{ad } g^{-1}(L \cap \text{Lie } P_g^-) \subset L$.

Denote by an index i the intersection with $G_i = \alpha(\pi^i L)$. Under hypotheses (2) and (3), there is i_0 such that, for $i \geq i_0$,

- (4) G_i is a subgroup of $G(k)$, and $G_i = U_{gi}^+ M_{gi} U_{gi}^-$;
- (5) $\text{Int } g(U_{gi}^+) \subset U_{gi}^+$, $\text{Int } g(M_{gi}) = M_{gi}$, and $\text{Int } g^{-1}(U_{gi}^-) \subset U_{gi}^-$.

This follows from the decomposition

$$U_g^+ \times M_g \times U_g^- \hookrightarrow G, \quad (u^+, m, u^-) \mapsto u^+ m u^-,$$

of a neighborhood of e .

Moreover, after increasing i_0 if necessary, there is a neighborhood V of g such that (2)–(5) remain true after replacing g by any $g' \in V$. For (2) and (3'), one uses the continuous dependence of P_g^+ and $\text{ad } g$ on g ; for (4) and (5), one uses the analytic dependence of m_g , and hence of $P_g^\pm$, on g , so that in the decomposition $x = u^+ m u^-$ the components u^+ , m , and u^- are analytic functions of the pair (x, g) .

Lemma. *Assume (1)–(5), let $i \geq i_0$, and suppose $\tau(g) \notin G_c^{\text{ad}}$. Then $\rho(G_i)\rho(g)\rho(G_i)$ is nilpotent.*

For the big-cell decomposition $U_g^+ \times M_g \times U_g^- \rightarrow G$, Haar measures satisfy $dg = \lambda(m) du^+ dm du^-$, where λ is a character of M_g . On every compact subgroup of M_g , $\lambda = 1$, so

$$[G_i] = [U_{gi}^+] \star [M_{gi}] \star [U_{gi}^-].$$

We prove by induction on N that

$$([G_i]g[G_i])^N = [G_i]g^N[G_i]$$

in the algebra of compactly supported distributions on G , writing g for δ_g . Indeed,

$$\begin{aligned} ([G_i]g^N[G_i])([G_i]g[G_i]) &= [G_i]g^N[G_i]g[G_i] \\ &= [G_i]g^N[U_{gi}^+][M_{gi}][U_{gi}^-]g[G_i] \\ &= [G_i][g^N U_{gi}^+ g^{-N}][g^N M_{gi} g^{-N}]g^N g[g^{-1} U_{gi}^- g][G_i] \\ &= [G_i]g^{N+1}[G_i]. \end{aligned}$$

If $\tau(g) \notin G_c^{\text{ad}}$, then the set of $\tau(g^N)$, $N \geq 1$, is not relatively compact. Since ρ is supercuspidal, there is an N such that

$$(\rho(G_i)\rho(g)\rho(G_i))^N = \rho(G_i)\rho(g^N)\rho(G_i) = 0.$$

Proof of the theorem. Let $g \in G(k)$ be such that $\tau(g) \notin G_c^{\text{ad}}$. Choose L , α , i_0 , and V as above, with $gG_{i_0} \subset V$. Let Θ be the character of ρ . For every $g' \in gG_i$ and every $i \geq i_0$, we have

$$\Theta(g'[G_i]) = \Theta([G_i]g'[G_i]) = \text{Tr}(\rho(G_i)\rho(g')\rho(G_i)) = 0$$

by the lemma. Since the subgroups G_i form a fundamental system of neighborhoods, Θ is zero on gG_i ; hence its support does not contain g . $\square$

Remarks. (1) The same theorem, with the same proof, holds for an analytic group that is a finite étale cover of an open subgroup of G . The hypothesis becomes: the image in G^{ad} of the support of the matrix coefficients is relatively compact. The conclusion becomes: the image in G^{ad} of the support of Θ lies in G_c^{ad} .

(2) Let p be the residue characteristic of k . The proof uses Haar measures $[K]$ in an essential way only for small K , hence in particular for a p -group. Therefore the theorem and its proof also hold for representations with coefficients in any field of characteristic $\neq p$.

Once a Haar measure dg on G is chosen, the character Θ of ρ is the distribution on G whose value on a test function φ is

$$\text{Tr} \left(\int \varphi(g) \rho(g) dg \right).$$

More intrinsically, it is the generalized function, i.e. the linear form on the space of measures $\varphi(g) dg$ with φ locally constant of compact support and dg a Haar measure, given by the same formula.

Institut des Hautes Etudes Scientifiques
Le Bois-Marie, 91440 Bures-sur-Yvette